Setting the Stage—Multimodality Treatment Approach in H&N Cancer: Current Standards, Challenges, and Unmet Needs

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Head and Neck Cancer

NPC

EBV

HPV

Non-NPC

SCCHN

Oral cavity
Oropharynx
Hypopharynx
Larynx

Other

Salivary gland
Nasal cavity
Paranasal sinus
Ear

EBV, Epstein-Barr virus; HPV, human papillomavirus; NPC, nasopharyngeal cancer; SCCHN, squamous cell carcinoma of the head and neck

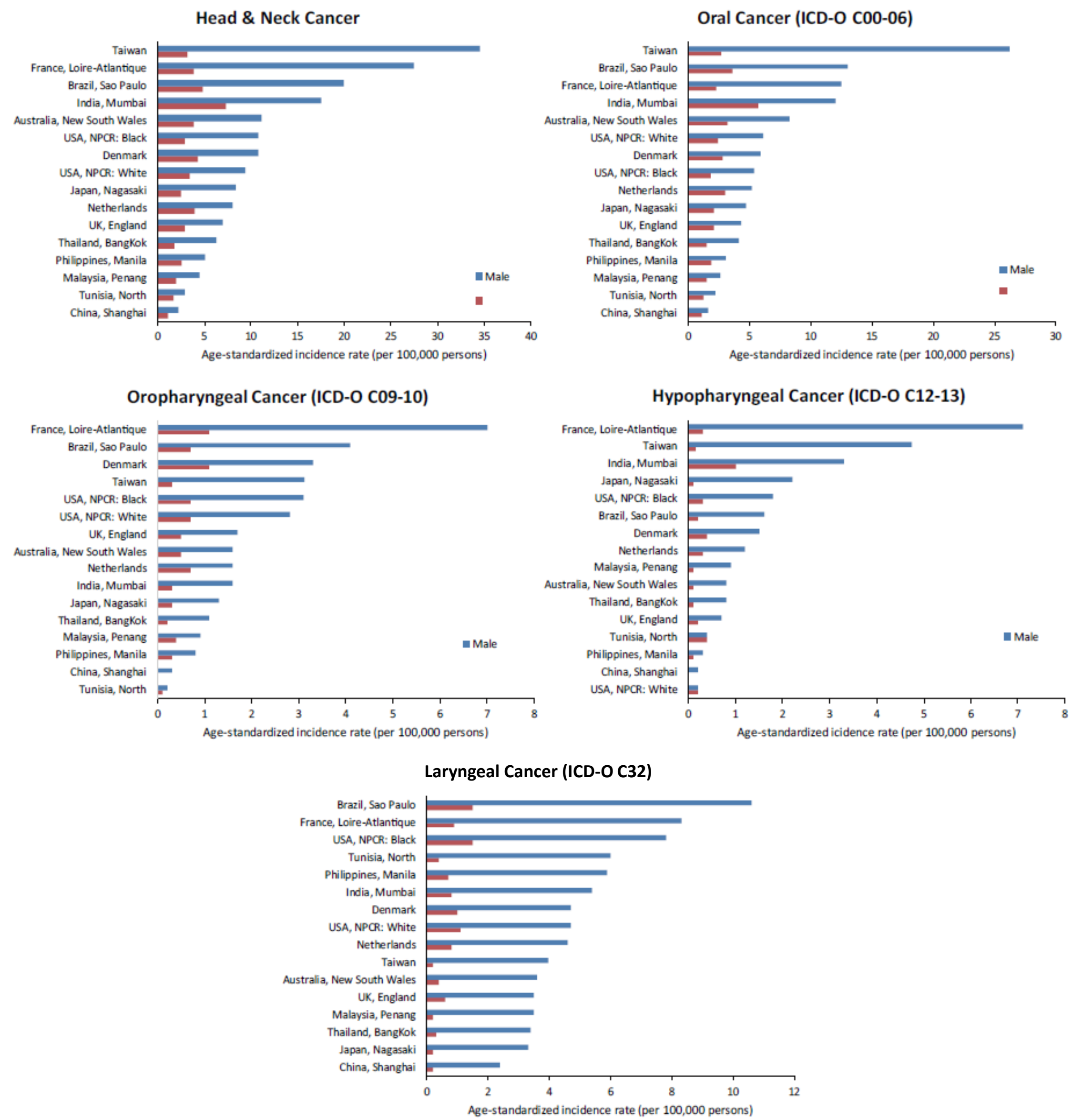
Epidemiology, Risk Factors, & Presentation

- 5% to 6% of all cancers (742,270 new cases in 2015; 407,037 deaths)
- ≥50% are in patients ≥60 years of age; >90% squamous cell CA type
- Risk factors:
  - Tobacco smoking
  - Alcohol use
  - Betel chewing
  - Virus (HPV-OPC, EBV-NPC)
  - Poor oral health
  - Mechanical irritation
  - Occupational exposure
  - Malnutrition
- Localized disease ~40%, regional metastases ~ 50%, distant metastases ~ 10%
- 2/3 locally/regionally advanced
- Major threat: Locoregional rec, SPT, SFT

OPC, oropharyngeal cancer; SFT, second field tumor; SPT, second primary tumor

2017 ESMO Essentials for Clinicans-Head and Neck Cancers. https://oncologypro.esmo.org/content/download/113096/1971597/file/2017-ESMO-Essentials-for-Clinicians-Head-Neck-Cancers.pdf. Accessed 6 December 2018. Braakhuis BJ, et al. *Oncologist.* 2005;10(7):493-500.

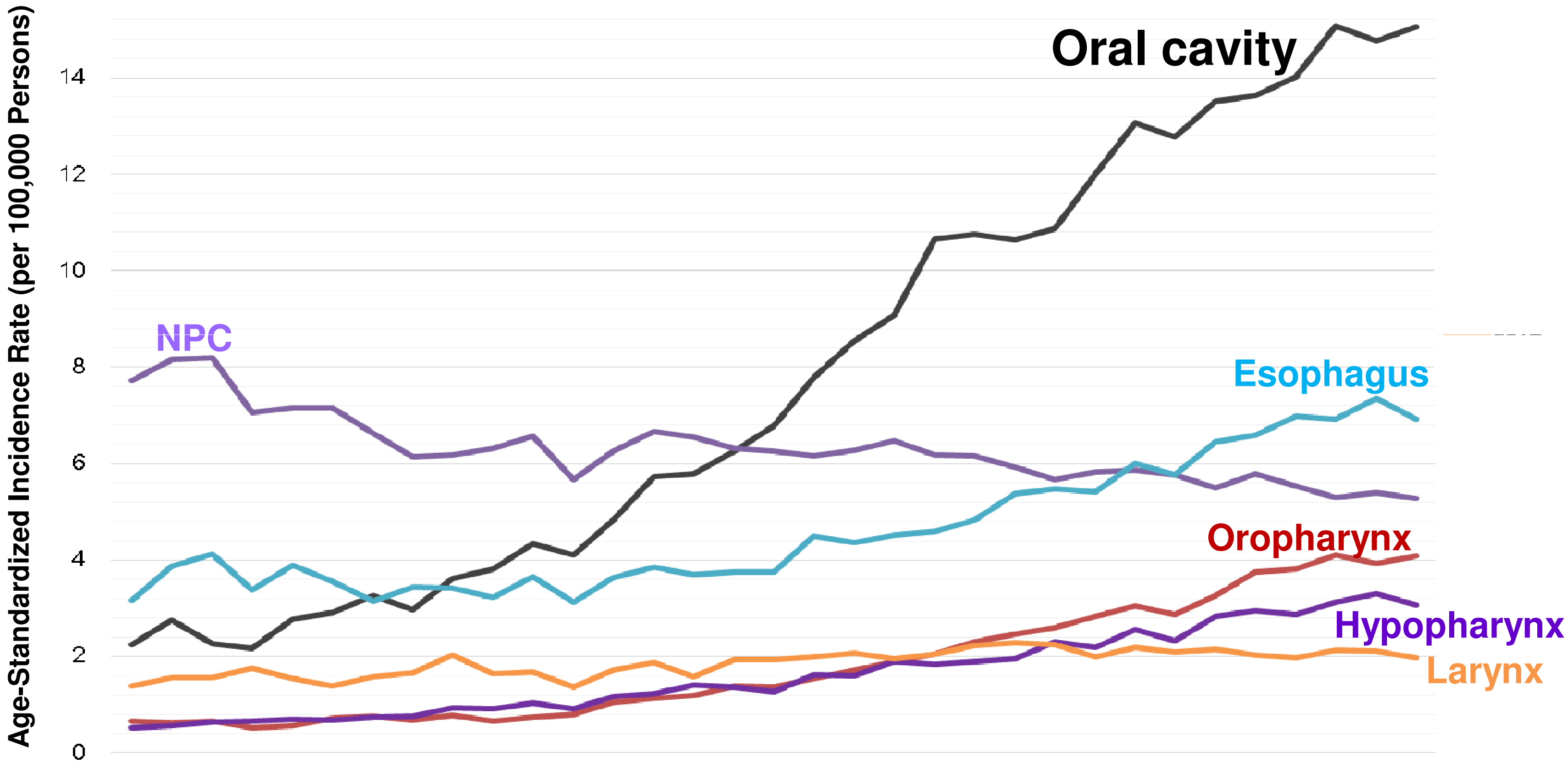
ASIR (per 100,000) of H&N Cancer and Subsites in Selected Countries, 2003 ~ 2007



ASIR, age-standardized incidence rate; H&N, head and neck

Hsu WL, et al. *Int J H&N Science*. 2017;1(3):180-189.

Age Standardized Incidence Rate of UADT Cancers in Taiwan, 1979-2011



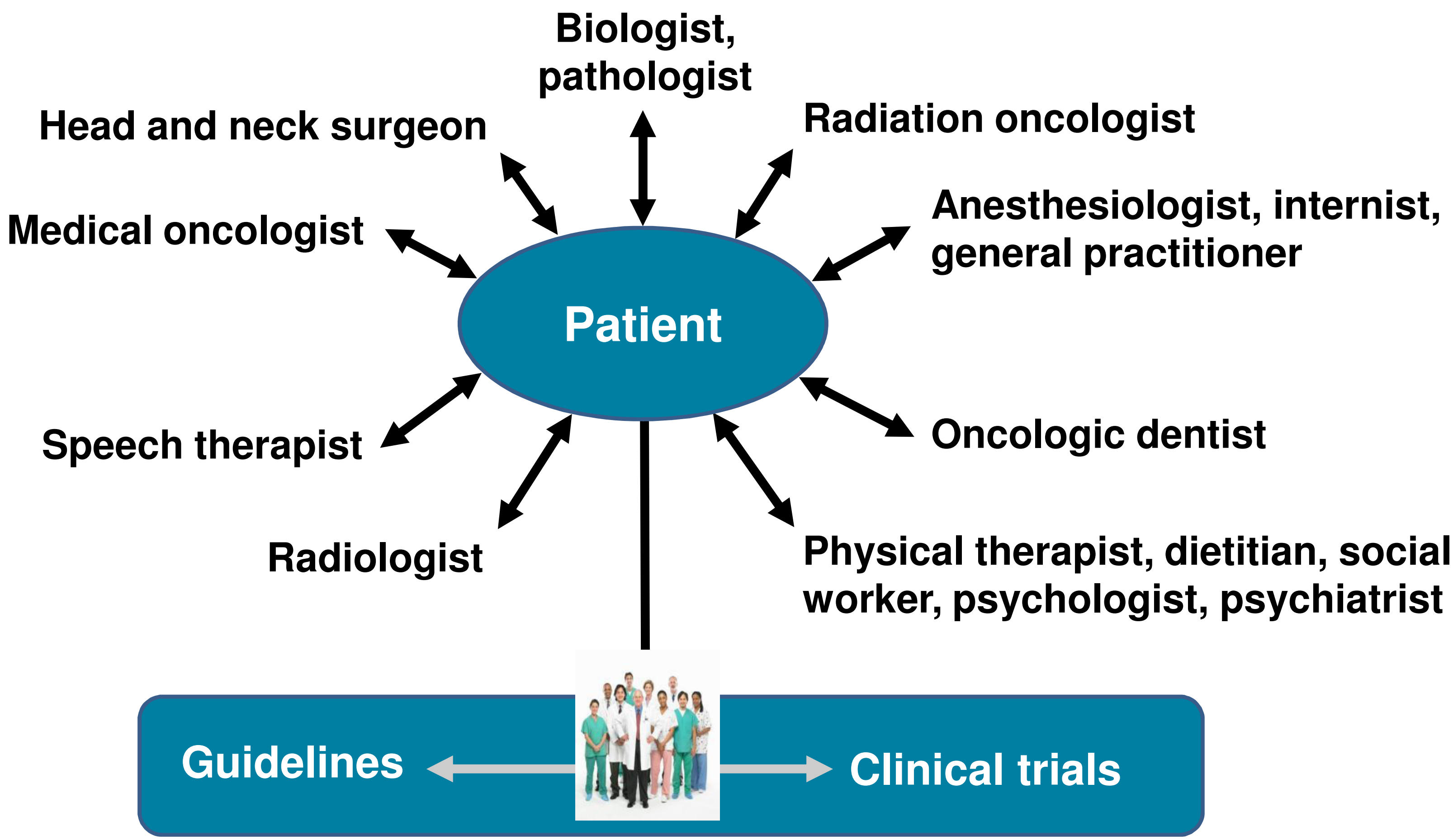
UADT, upper aerodigestive tract

Taiwan Cancer Registry, 2014. http://tcr.cph.ntu.edu.tw/main.php?Page=N2. Accessed 7 December 2018.

Characteristics of SCCHN

1. A complex disease entity
   – Complexity in anatomy & genetics
2. Diversity of biological behavior
3. Comorbidity, impaired vital functions
4. Multiple primaries
5. Needs a multidisciplinary team (MDT)

MDT Meetings

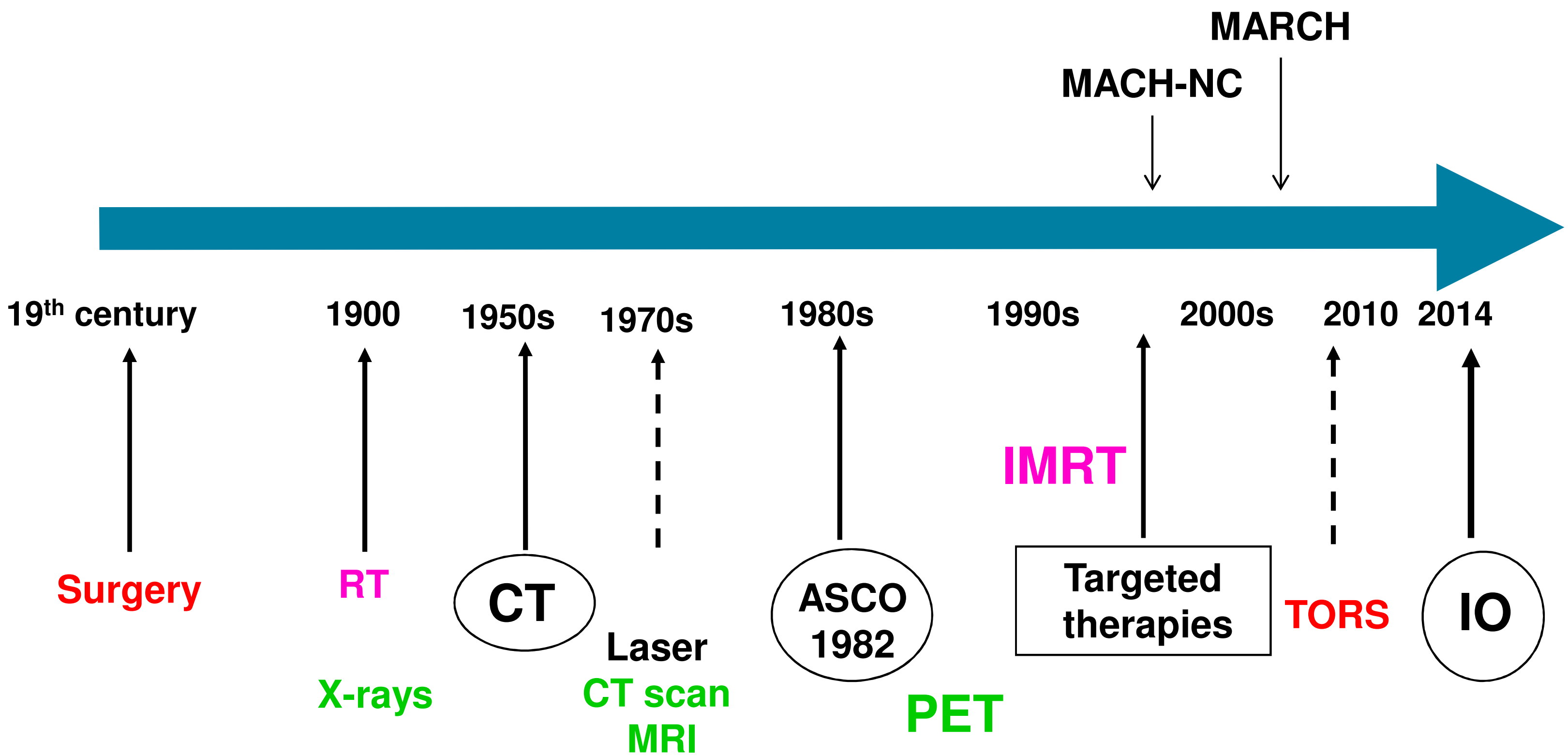


Benefits of the MDT

1. Improved survival
2. Improved staging accuracy
3. Improved adherence to clinical practice guidelines
4. More cost-effective care
5. Better communication leading to clinician and patient satisfaction

Licitra L, et al. *Oral Oncol.* 2016;59:73-79.

Milestones in H&N Cancer Management



ASCO, American Society for Clinical Oncology; CT, chemotherapy; CT scan, computed tomography scan; IO, immuno-oncology; IT, immunotherapy; MACH-NC, meta-analysis of chemotherapy in head and neck cancer; MARCH, meta-analysis of radiotherapy in squamous cell carcinomas of head and neck; MRI, magnetic resonance imaging; PET, positron emission tomography; RT, radiotherapy; TORS, Trans-oral robotic surgery

Meta-analysis of Locoregional Treatment With or Without Chemotherapy (Effect on OS)

Trial Category	HR (95% CI)	C/T effect (*P*-value)	Heterogeneity (*P*-value)	Absolute benefit	
				2-year	5-year
AdjCT	0.98 (0.85-1.19)	.74	0.35	1%	1%
NeoCT	0.95 (0.88-1.01)	.10	0.38	2%	2%
ConCT	0.81 (0.76-0.88)	<.0001	<0.0001	7%	8%
Total	0.90 (0.85-0.94)	<.0001	<0.0001	4%	4%

63 trials, 10741 patients, cancer of oropharynx, oral cavity, larynx, hypopharynx

UPDATE:

- 24 new trials, total 16,665 patients, overall HR=0.88 (*P*<.0001), 4.4% at 5-yr
- More significant for CCRT: HR=0.81 (*P*<.0001), 8% at 5-yr

Pignon JP, et al. *Lancet.* 2000:355(9208):949-955. Pignon JP, et al. Int J Radiat Oncol Biol Phys. 2007;69(2Suupl):S112-114.

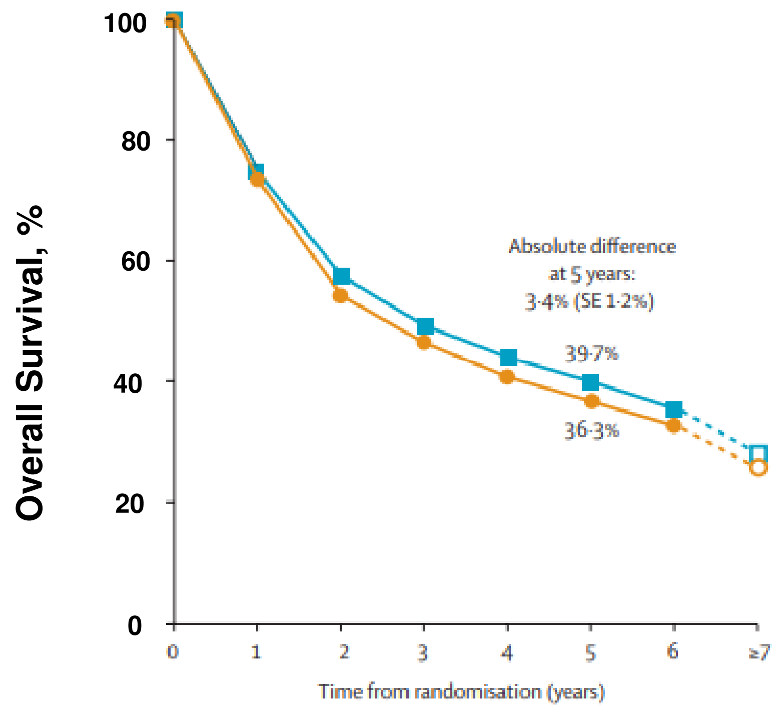
MACH-NC Update: Indirect Comparison of NeoCT vs ConCT

Type of Chemotherapy	HR		Interaction test R=HR	
	CCRT vs RT	NeoCT-RT vs RT	P	CCRT/NeoCT-RT
Platinum + 5FU	0.77 (0.69-0.85)	0.90 (0.82-0.99)	.03	0.86
Other polyCT	0.80 (0.71-0.91)	1.01 (0.91-1.12)	.0042	0.80
MonoCT platinum	0.74 (0.67-0.82)	No trial	-	-
MonoCT other	0.89 (0.82-0.96)	0.99 (0.84-1.18)	.25	0.90
Total	0.81 (0.78-0.86)	0.96 (0.90-1.02)	.0001	0.84

- Direct Comparison of CCRT vs NeoCT-RT (6 trials, 861 patients)
- Better effect of CCRT, but not significant (HR=0.90, 95%CI 0.77-1.04, *P*=.15)

Pignon JP, et al. Int J Radiat Oncol Biol Phys. 2007;69(2Suupl):S112-114.

Altered vs Conventional Fractionation RT



Death/person years by period	Years 0–2	Years 3–5	Years ≥6
Conventional RT	1548/4988	506/3729	181/1643
Altered fractionated RT	1526/5447	563/4372	224/2042

Bourhis J, et al. *Lancet.* 2006:368(9538):843-854.

Variation of Treatment Effect According to Age in the CCRT Part of MACH-NC And MARCH

Age	Altered Fr RT vs Standard RT			CCRT vs RT		
	Cases	HR (95% CI)	Test for trend	Cases	HR (95% CI)	Test for trend
≤50	1311	0.78 (0.65-0.94)	.007	2584	0.76 (0.66-0.86)	.003
51-60	2300	0.95 (0.83-1.09)		3306	0.78 (0.70-0.87)	
61-70	2346	0.92 (0.81-1.06)		2698	0.88 (0.78-1.00)	
≥71	1085	1.08 (0.89-1.30)		692	0.97 (0.76-1.23)	

- OS effect decreased with increased age
- Deaths not due to HNC increased with age, from 18% (50 yr) to 41% (≥71 yr) in MARCH, and from 15% to 39% in MACH-NC studies

Pignon JP, et al. Int J Radiat Oncol Biol Phys. 2007;69(2Suupl):S112-114.

Treatment Classification

Surgery (Local effect)

Nonsurgical Treatment

- Radiotherapy (Local effect)
- Drugs (Systemic + Local effect)
  - Chemotherapy
  - Targeted therapy, IO therapy

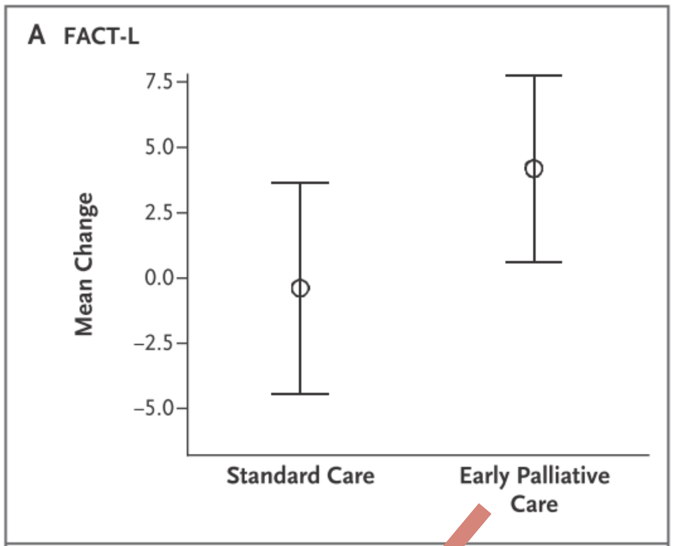
Palliative care (QoL)

MDT is the key!

QoL, quality of life

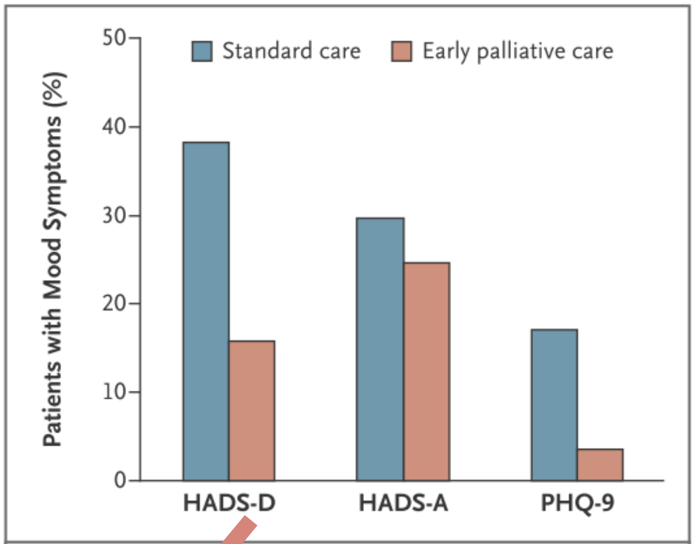
Importance to Early Integrate Palliative Care With Standard Oncologic Care

Mean Change in QoL Scores From Baseline to 12 Weeks



Higher FACT-L score
98.9 vs 91.5, *P* = .03
Better QoL

12 Week Outcomes of Assessments of Mood

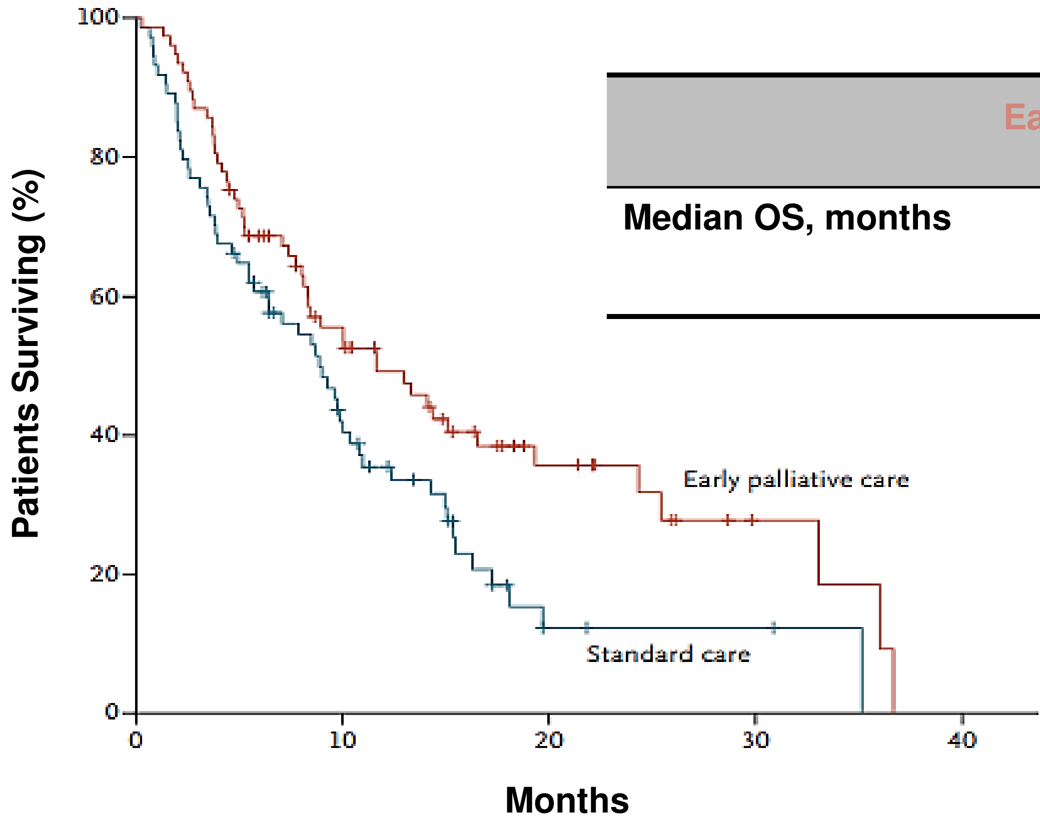


Less depressive symptoms
16% vs 38%, *P* = .01

Temel JS, et al. *N Engl J Med.* 2010;363(8):733-742.

FACT-L, Functional Assessment of Cancer Therapy–Lung; HADS, Hospital Anxiety and Depression Scale; HADS-A, HADS-anxiety; HADS-D, HADS-depression; PHQ-9, Patient Health Questionnaire 9

Improved Survival With Early Integration of Palliative Care in Cancer Treatment

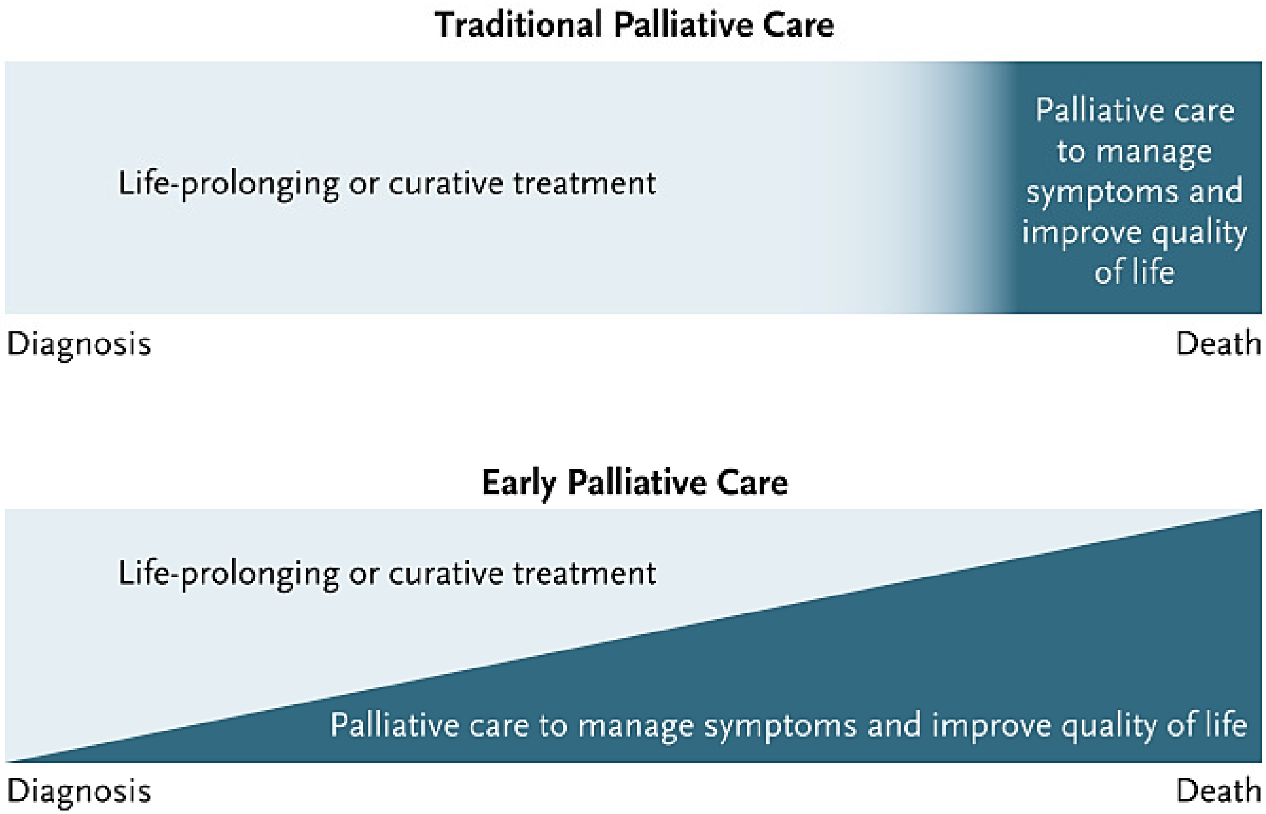


	Early Palliative Care (N = 77)	Standard Care (N = 74)
Median OS, months	11.6	8.9
	P = .02	

OS, overall survival

Temel JS, et al. *N Engl J Med.* 2010;363(8):733-742.

Traditional vs Early Palliative Care



Parikh RB, et al. *N Engl J Med.* 2013;369(24):2347-2351.

Surgery

Local effect

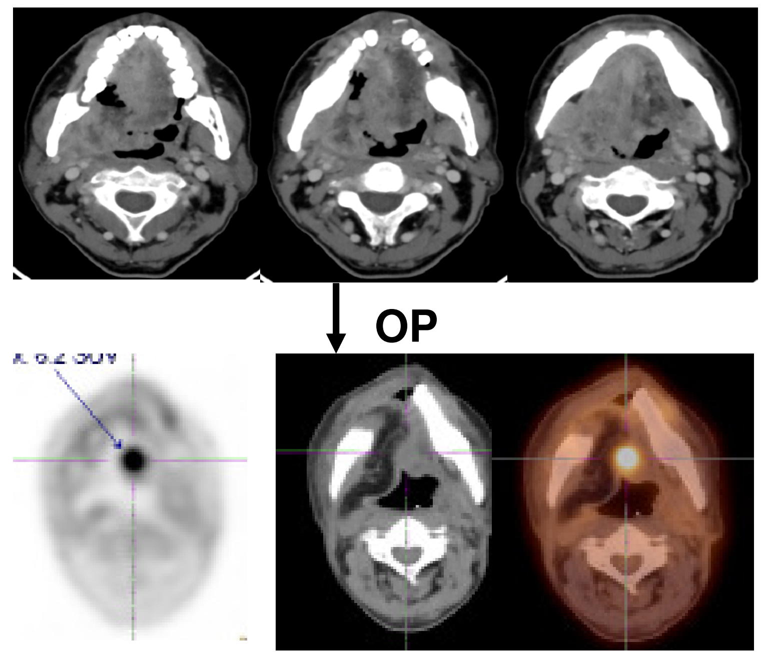
Rapid efficacy

Not resistant but

- undetectable cells
- failed in margin

Need adjuvant RT-CT

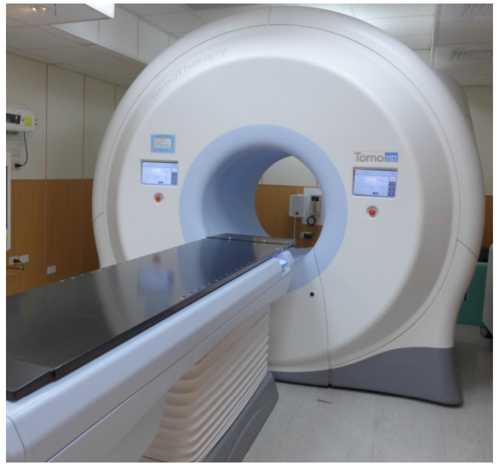
- for LA SCCHN



Op, operation; LA SCCHN, locally advanced squamous cell carcinoma of the head and neck

Images courtesy of Lin JC.

Radiotherapy



- Local effect
- Long Tx time, slow regression
- Acute & late toxicities
- Inadequate for LA SCCHN
- Radioresistance problem

Tx, treatment

Milestones of Systemic Therapies in SCCHN

1960s	**Single-agent chemotherapy** Methotrexate
1970s	**Combination chemotherapy regimens** Platinum compounds
1980s	**Induction chemotherapy** Larynx preservation
1990s	**Concurrent chemoradiotherapy (CCRT)** Taxanes
2000s 2010s	**Induction chemotherapy revisited** Targeted therapy (cetuximab) Immunotherapy

Chemotherapy

- Established efficacy
- Systemic and local effects
- Drug resistance
- Less durable response in solid tumors
- Associated with certain morbidity

Targeted Therapy (Cetuximab)

- Systemic and local effects
- Established efficacy
- Variation in response duration
- Different toxicity compared with CT

IO Therapy (ICI)

- Systemic and local effects
- Established efficacy
- Durable remission in some cases
- Delayed clinical effect
- Continue treatment beyond progression (selected patients)
- Less toxic (monitoring for irAEs required)

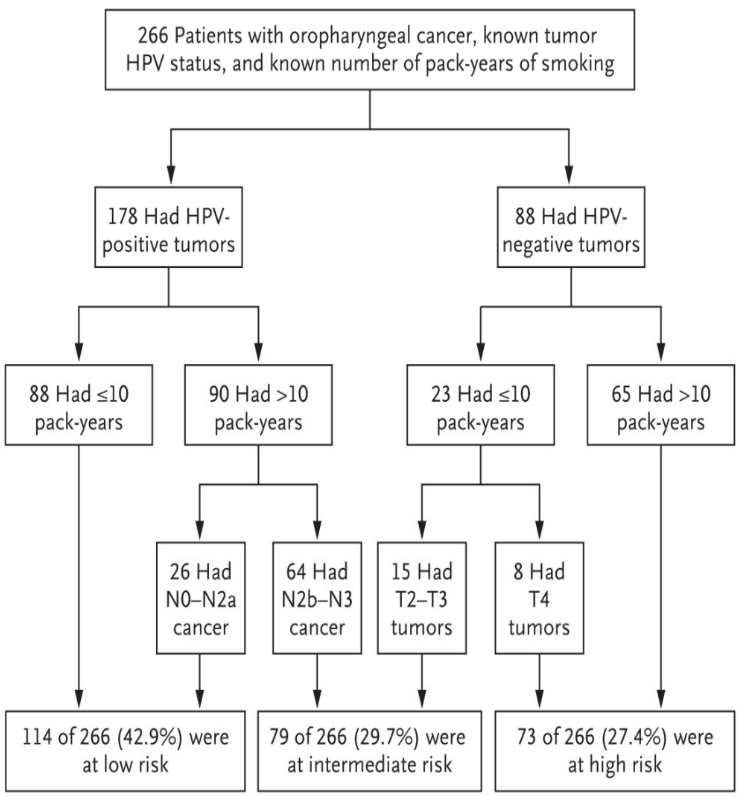
ICI, immune checkpoint inhibitors; irAE, immune-related adverse event

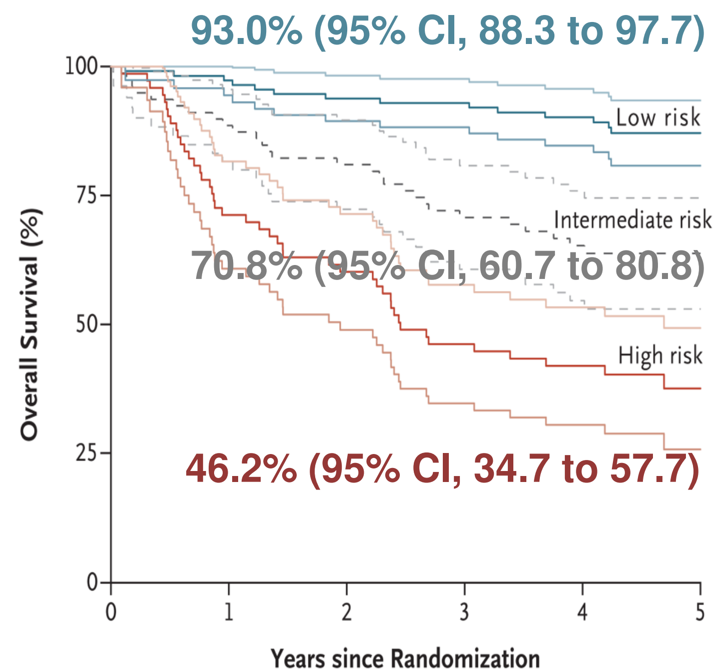
Important Recent Findings

- HPV is a risk factor for OPC (a growing epidemic)
- Tumor HPV single strongest predictor of survival (OPC)
- EGFR is a second prognostic marker
- Anti-EGFR medication is getting major attention
- Expanded role of chemotherapy (CCRT, ICT)
- Revival of immunology/immunotherapy (ICI)
- Improved RT techniques (IMRT, IGRT, proton)
- Functional imaging (PET/CT, PET/MR)
- QoL of survivors is getting more attention

EGFR, epidermal growth factor receptor; IGRT, image guided radiation; IMRT, intensity modulated radiotherapy

The Prognostic Significance of HPV in OPC





No. at Risk

Low risk	114	111	106	102	95	46
Intermediate risk	79	70	64	54	44	24
High risk	73	52	43	33	28	8

Ang KK, et al. ***N Engl J Med.*** **2010;363(1):24-35.**

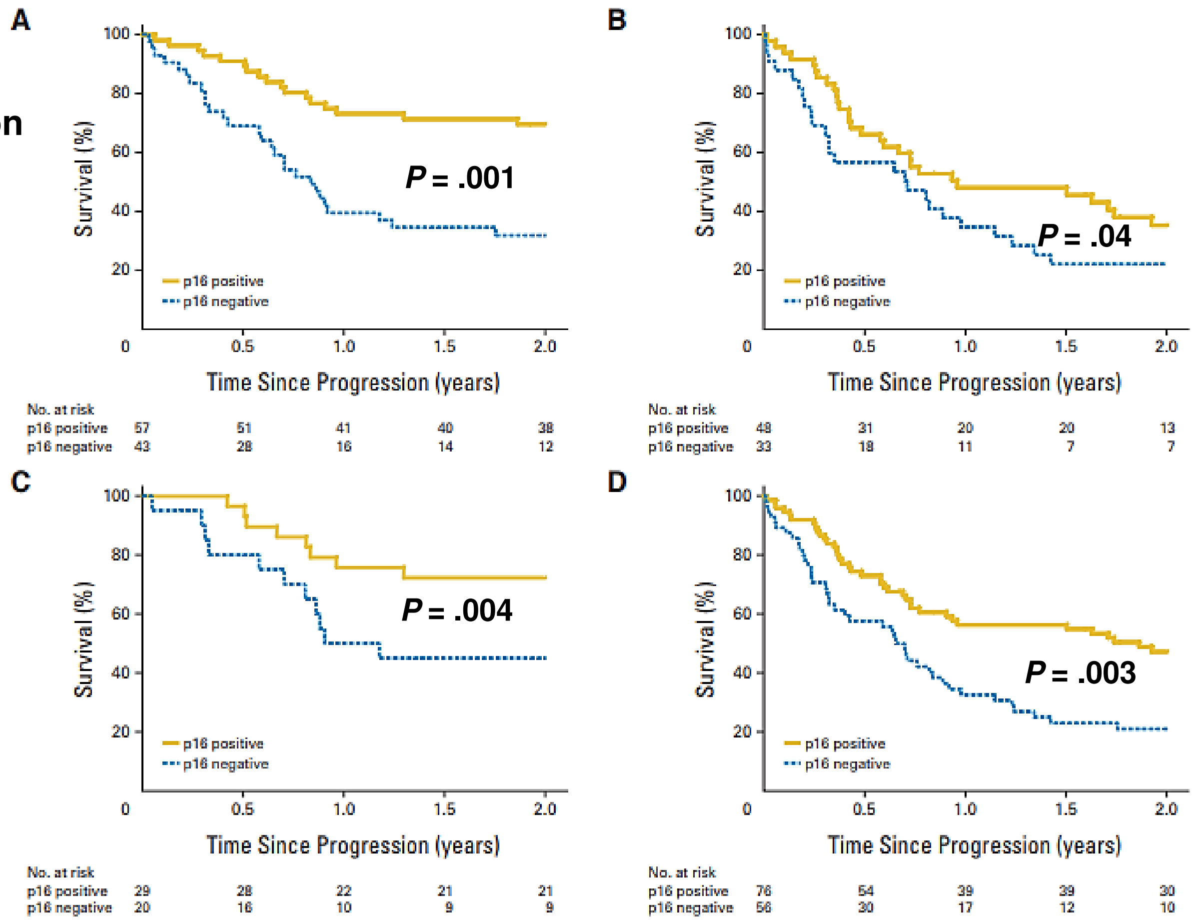
Impact of p16 Status on Overall Survival After Disease Progression in OPC

A. Patients with locoregional progression

B. Patients with distant progression

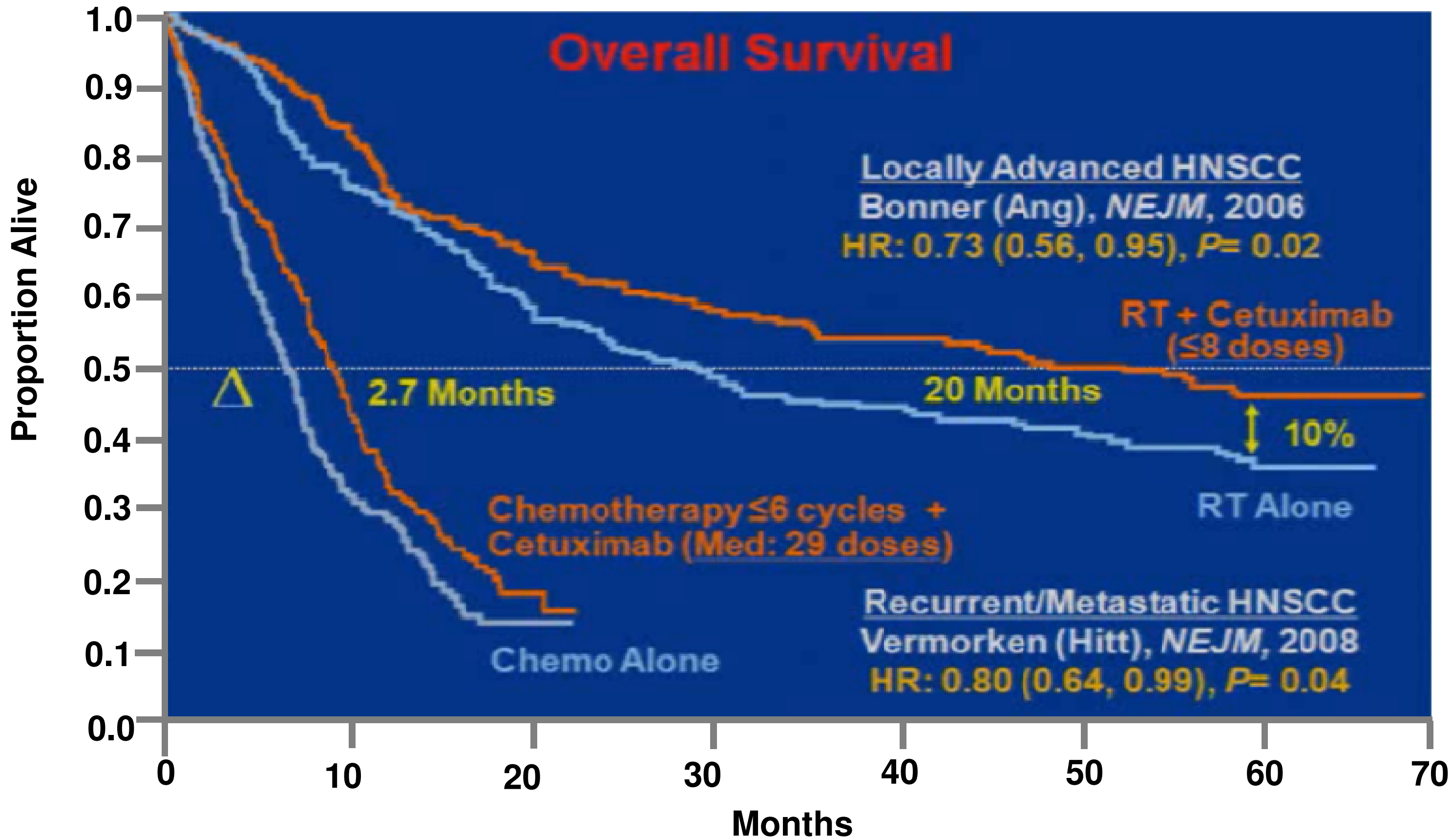
C. Patients who had salvage surgery

D. Patients with no salvage surgery



Fakhry C, et al. *J Clin Oncol.* 2014;32(30):3365-3373.

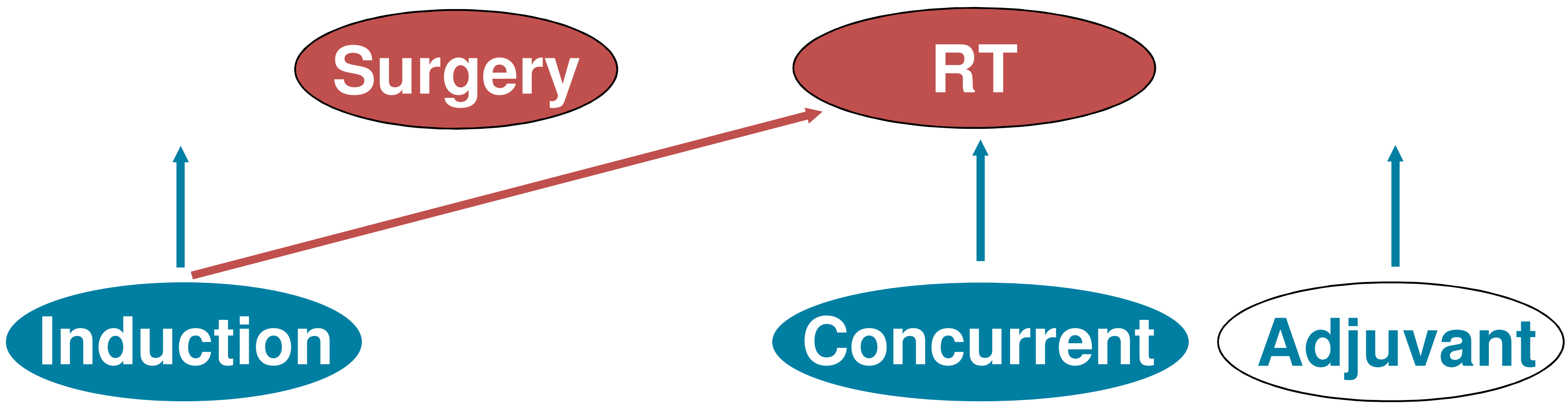
Combining Cetuximab With Radiotherapy or Platinum-Based Chemotherapy Improves Survival



Bonner JA, et al. *N Engl J Med.* 2006;354(6):567-578. Vermorken JB, et al. *N Engl J Med.* 2008;359(11):1116-1127.

Treatment Considerations for LA SCCHN (MDT)

Local Treatment



Systemic Tx (CT, Targeted Tx, IO)

Guidelines as a reference + individualized treatment

Unmet Needs of Locally Advanced SCCHN

After initial MDT treatment

- **CR → a high relapse rate**
- **PR/SD → progress rapidly**
- **PD → die soon**

Cure → Late morbidity & 2^{nd} primary

CR, complete response; PD, progressive disease; PR, partial response; SD, stable disease

Treatment Considerations for Recurrent/Metastatic (R/M) SCCHN

Local therapy feasible → Op/RT/CCRT

Unsuitable → Drug therapy

CT (PF, MTX,…)

Targeted treatment (cetuximab,…)

EXTREME

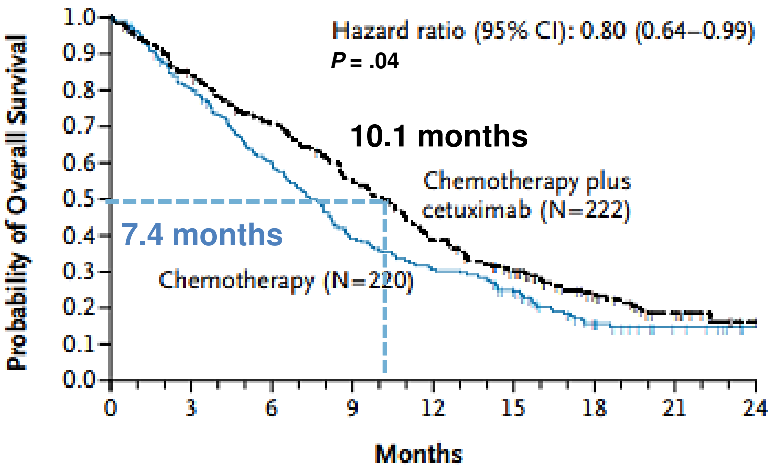
IO (nivolumab/pembrolizumab,…)

Clinical trials

(Efficacy/toxicity/cost of each modality, patient willing)

MTX, methotrexate; PF, +5FU

EXTREME Was the First Trial Showing OS Benefit in R/M SCCHN



- PFS = 5.6 vs 3.3 months (*P*<.001)
- Time to Tx failure = 4.8 vs 3.0 months (*P*<.001)
- Of those patients treated with platinum-based therapy, 11% were still alive after 2 years

	N	Regimen	ORR, %	CR, %	mPFS, Months	mOS, Months	*P* Value (mOS)
Wittes, 1977	26	· Cisplatin	31	7.7	NR	NR	NR
Forastiere, 1992	277	· Cisplatin + 5FU · Carboplatin + 5FU · Methothrexate	32 21 10	6.0 2.0 2.0	NR	6.6 5.0 5.6	NS
Jacobs, 1992	249	· Cisplatin + 5FU · Cisplatin 5FU	32 17 13	6.3 3.6 2.4	2.4 2.0 1.7	5.5 5.0 6.1	.49
Clavel, 1994	382	· CABO · Cisplatin + 5FU · Cisplatin	34 31 15	9.5 1.7 2.5	4.75 4.25 3.0	7.3 7.3 7.3	.11
Gibson, 2005	218	· Cisplatin + 5FU · Cisplatin + paclitaxel	27 26	6.7 7.0	NR	8.7 8.1	NS
Vermorken, 2008	442	· Cis/carboplatin + 5FU · Cis/carboplatin + 5FU + cetuximab	20 36	NR	3.3 5.6	7.4 10.1	$\boxed{.04}$

- RR = 36% vs 20% (*P*<.001)
- DCR = 80% vs 61% (*P*<.001)

DCR, disease-control rate; PFS, progression-free survival; RR, relative risk

Vermorken JB, et al. *N Engl J Med.* 2008;359(11):1116-1127. Vermorken JB, et al. *J Clin Oncol.* 2014;32(5s): Abstract 6021.

EXTREME Is Now Challenged by Immunotherapy

Trial	Experimental arm(s)		Standard of care
CheckMate 651[1] **N = 700** Primary endpoint: OS, PFS Recruiting	**Nivolumab + ipilimumab**	vs	Standard of care **EXTREME** Cis-/Carboplatin + 5FU + Cetuximab 6 cycles q 3 w followed by cetuximab maintenance
KEYNOTE-048[2] **N = 825** Primary endpoint: PFS Reported at ESMO	**Arm1: Pembrolizumab** **Arm 2: Pembro + platinum + 5FU**		
KESTREL[3] **N = 823** Primary endpoint: OS, PFS Accrual completed	**Arm1: Durvalumab** **Arm 2: Durva + tremelimumab**		

ESMO, European Society for Medical Oncology; Pembro, pembrolizumab

1. National Institutes of Health. https://clinicaltrials.gov/ct2/show/NCT02741570. Accessed 30 November 2018. 2. Burtness B, et al. *Ann Oncol.* 2018;29(Suppl 8): Abstract LBA8_PR. 3. National Institutes of Health. https://clinicaltrials.gov/ct2/show/NCT02551159. Accessed 6 December 2018.

KEYNOTE-048: First-Line Pembrolizuamb vs EXTREME and EXTREME vs Pembrolizumab + Chemotherapy in SCCHN

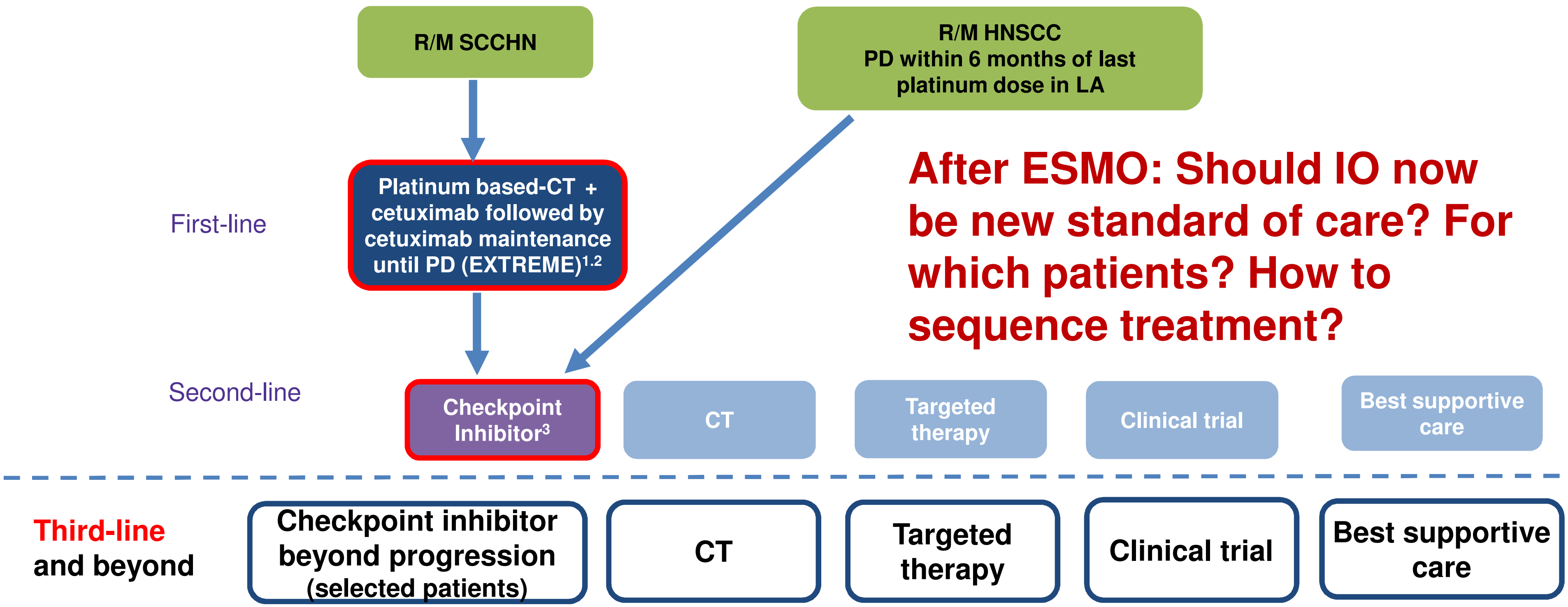
Results Summary

	Pembrolizumab (CPS ≥20)	EXTREME (CPS ≥20)		Pembro + Chemo
ORR, %	23.3	36.1	36.3	35.6
mDOR, months	20.9	4.2	4.3	6.7
mPFS, months	3.4	5.0	5.1	4.9
mOS, months	14.9	10.7	10.7	13.0
Grade 3-5 AE, %	16.7	69.0	69.0	71.0
All AE, %	58.3	96.9	96.9	95.3
Death, %	1.0	2.8	2.8	3.6

CPS, combined positive score; m, median

Burtness B, et al. *Ann Oncol.* 2018;29(Suppl 8): Abstract LBA8_PR.

Treatment Options for R/M HNSCC Before ESMO 2018



After ESMO: Should IO now be new standard of care? For which patients? How to sequence treatment?

1. National Comprehensive Cancer Network (NCCN) Clinical Practice Guidelines in Oncology: Head and Neck Cancer, V.2.2017. www.nccn.org/professionals/physician_gls/f_guidelines.asp. Accessed 6 December 2018. 2. Grégoire V, et al. *Ann Oncol.* 2010(Suppl 5);21:v184-v186. 3. Ferris RL, et al. *N Engl J Med.* 2016;375(19):1856-1867.

Unmet Needs in R/M SCCHN

- Low response rate
- Short duration of response (is sequence Tx better?)
- Toxicity
- Cost
- Clinically useful biomarkers

Thank You for Your Attention!